



Frame Extraction and Labelling of Social Media Posts

Nimra Nayyar, Levi Bates, Ameya Chaudhari, Ruoyu Yang, Fanchen Meng
Watt AI Program, Clemson University



Background

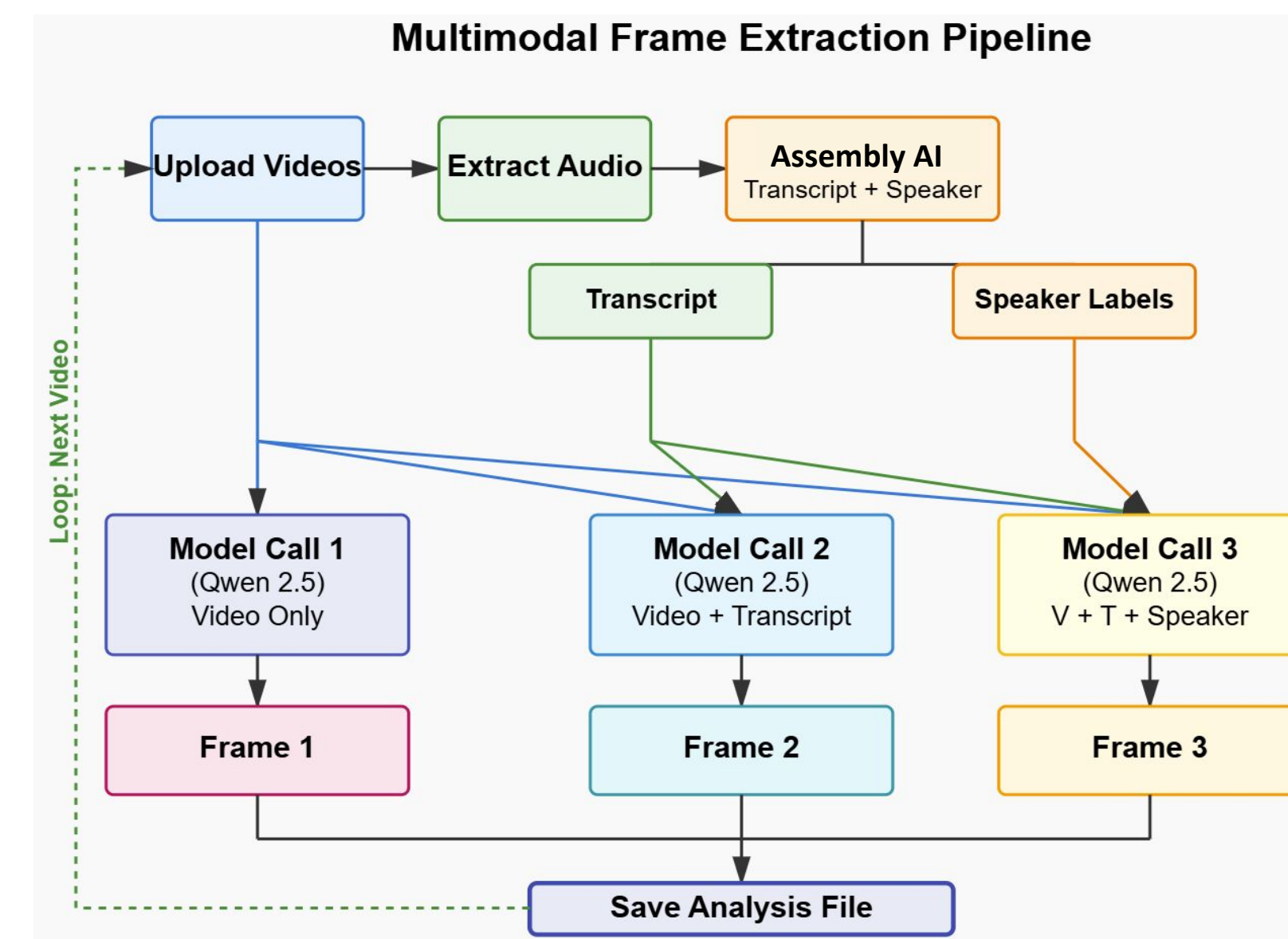
This project investigates how meaning and influence are constructed across social media text and video by modeling what core message audiences are meant to take away. It focuses on three related tasks: extracting the main idea from short-form videos using transcripts and video-aware LLMs, standardizing noisy or inconsistent person names into coherent entities, and building user-level profiles that summarize what each account is trying to communicate over time.



By combining transcript processing, entity clustering, and embedding-based user modeling, the project aims to turn raw social media content into structured representations of messages, people, and communities. This supports deeper analysis of online narratives, coordination, and information flows, while providing hands-on experience with LLMs, video models, embeddings, clustering, and dimensionality reduction tools.

Multimodal Video Frame Extraction

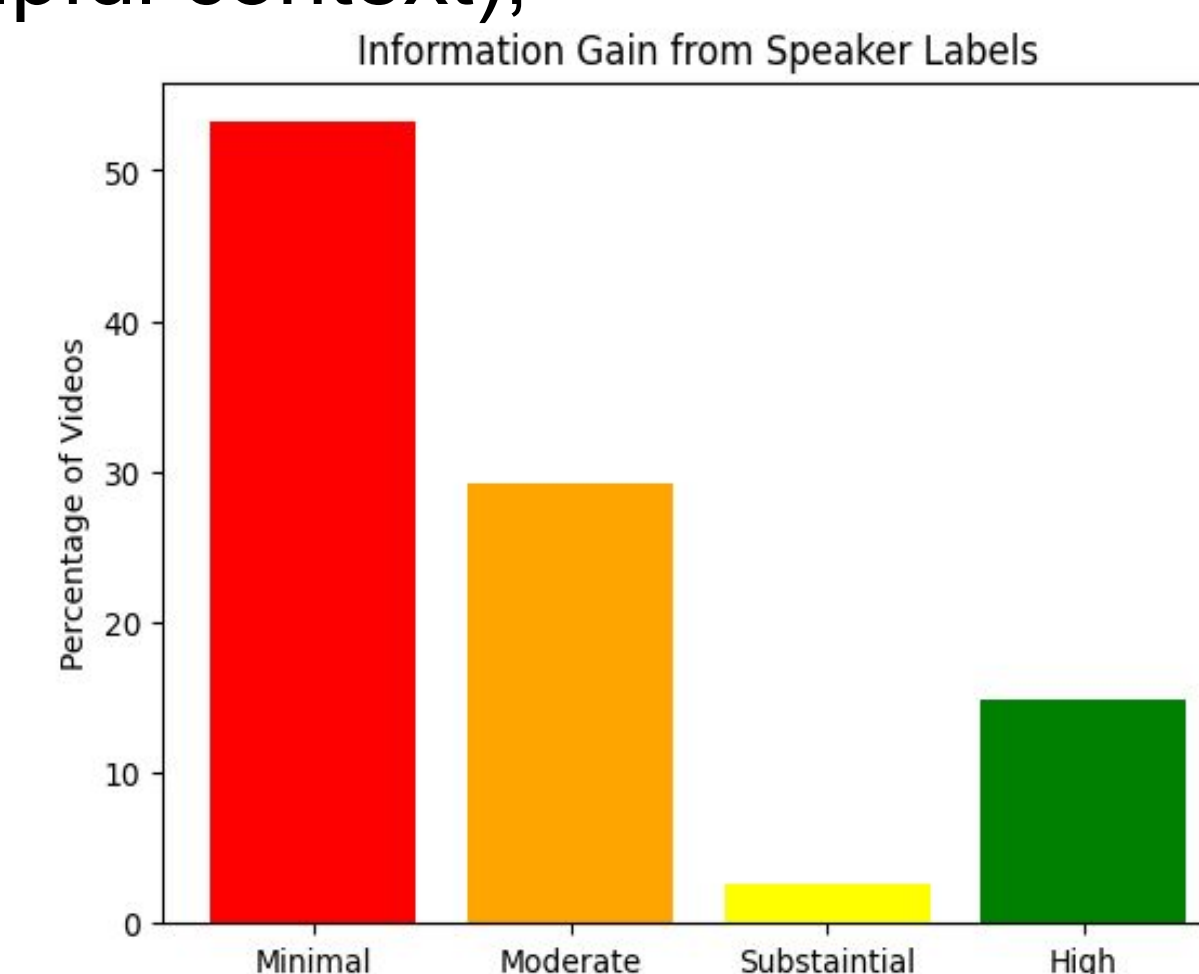
This project investigates how well vision-language models understand the core message of short-form social media videos. We built a multimodal pipeline that analyzes videos under three conditions: visuals only, visuals +transcript, and visuals +transcript +speaker labels. By comparing these outputs, we measure how much information each modality contributes and when speaker attribution meaningfully changes the model's interpretation



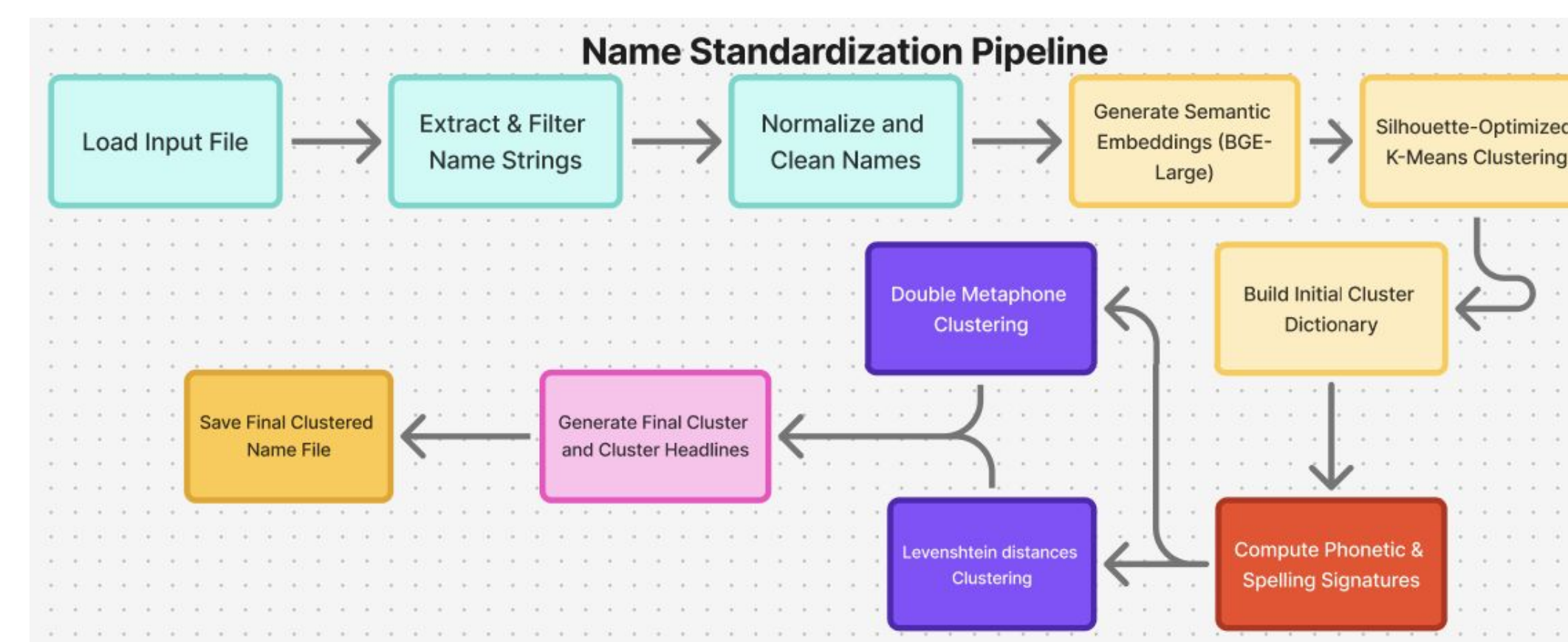
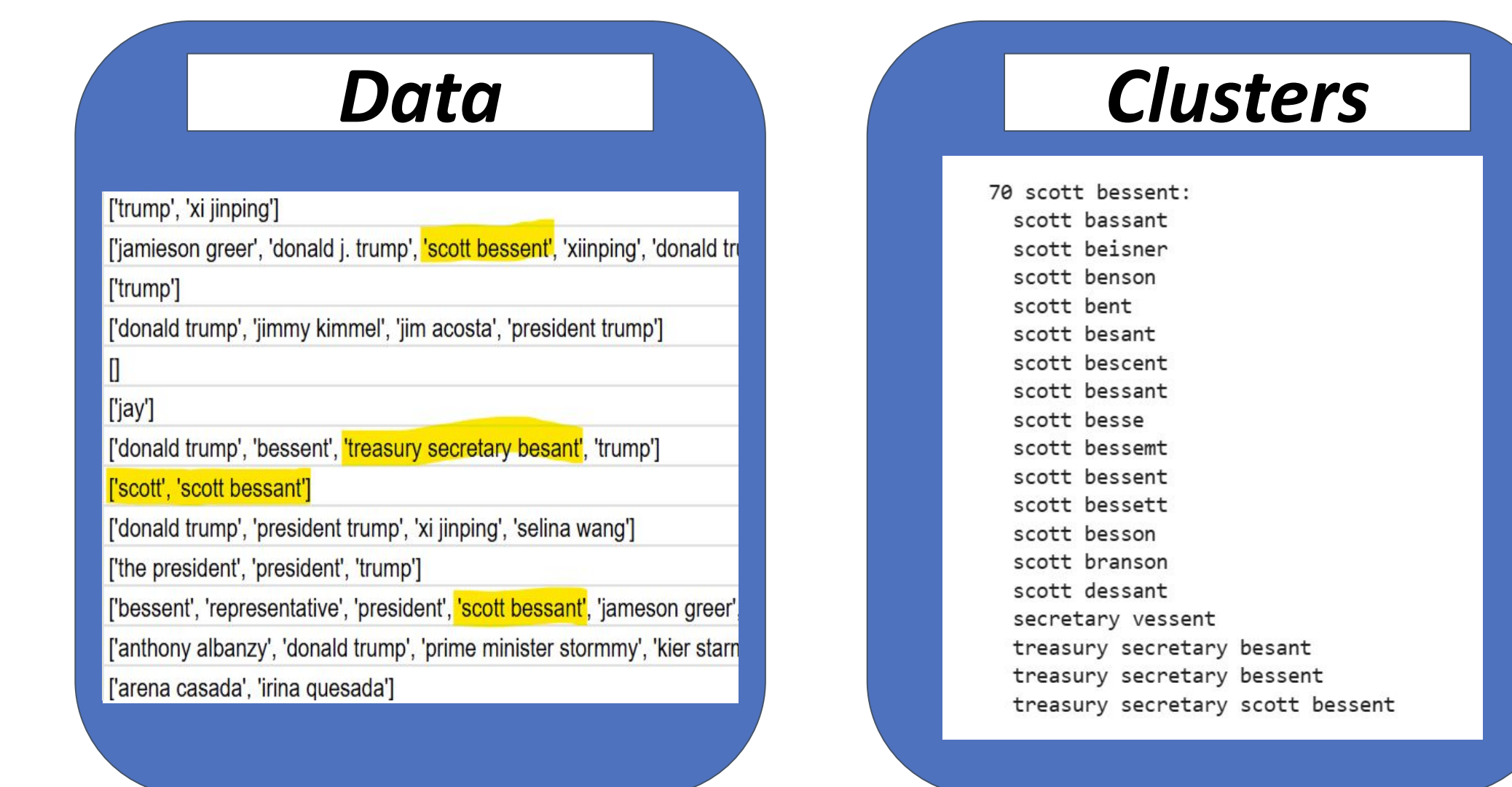
Information Gain Evaluation

We computed information gain by comparing semantic similarity between video+transcript and video+transcript+speaker labels using sentence embeddings (allMiniLM-L6v2 model). Results across 154 videos: 53.2% showed Minimal Gain (<15% gain, speaker labels redundant), 29.2% Moderate gain (15-30% gain, helpful context),

2.6% Substantial gain (30-50% gain, important information), and 14.9% High Gain (>50% gain, critical for understanding).



Name Standardization + Clustering



Embeddings and Pronunciation

The BGE Large Embedding model was used to transform each name into a high dimensional vector, allowing semantically similar names to appear close together. Pronunciation similarity and Levenshtein distance refined the results by handling types, abbreviations, and title variations.

Embeddings (Hugging Face BGE-Large)

- Generated vector representation of strings for context
- Clustered using Cosine similarity

Pronunciation (Double Metaphone)

- Created phonetic codes to catch misspellings/soundalikes
- Used Levenshtein to merge pronounced names

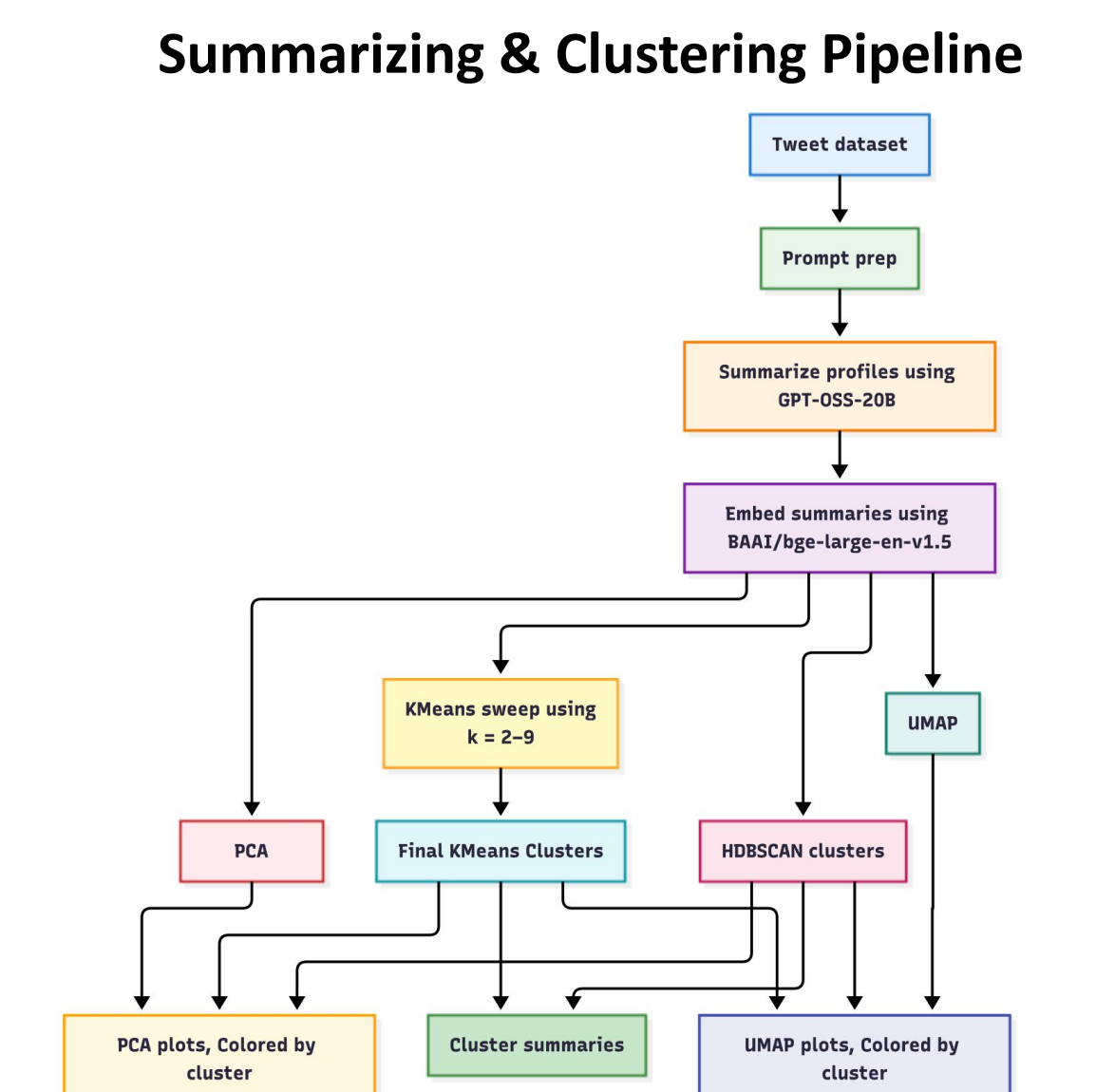
Final Merge

- Embedded Clusters formed initial group; Within each cluster pronunciation is used to detect errors
- Small clusters were emerged when semantic and phonetic similarity agreed.



Using LLMs to summarize and cluster users:

To automate objective user-level profiling, we used gpt-oss-20b to summarize each user's tweets. We then embedded these summaries using bge-large-en-v1.5 and looked for clusters in the resulting vectors. We then compared these summaries and resulting clusters with human thematic groupings to assess LLM effectiveness.



Future Work

- Integrate LLMs with reinforcement learning from human feedback to optimize performance further